

GIS User Guide

Print Server 2.0

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Notifications are made throughout this document for the following reasons:

Safety



Information alert that can prevent possible damage to the user, product or other objects



Information caution that can avoid possible damage to the user, product or other objects

Information



Notes provide added information for the user



Recommendations indicate advised procedures or maintenance in order to maximise product functionality and lifespan

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1 Introduction

The purpose of this manual is to provide the necessary information for using the features of the **GIS Print Server 2**, such as the interface and configuration management.

- This manual does not cover the functionality of the Engine Components, such as the Render Engine and Print Controller - for further information, see the appropriate Engine Component User Guide
- This manual does not cover the contents of the individual configuration files - for more information, see the appropriate Engine Component Configuration Reference
- This manual does not cover TCP/IP communications with the Print Server - for more information, see the *Print Server Communication User Guide 2.0*

2 Starting Print Server

To start using the GIS Print Server 2 software application, double click the "Launch GIS Print Server 2" icon  on the desktop.

If for any reason the icon is not there, you can launch the application from the relevant drive in:

C:\Program Files\Global Inkjet Systems\GIS Inkjet OS 2\GIS Print Server 2.exe

2.1 Launching Through the Command Line

Print Server 2 can be launched through the command line with optional start up parameters. The command for launching Print Server 2 with parameters is:

"GIS Print Server 2.exe" "<ID>,<Print Server Port>:<Binary Data Port>,<Configuration Folder>,<Start Minimised>"

<ID>	This is the ID of the Print Server 2. This is useful for differentiating between multiple Print Servers that may be open at the same time, as the ID is displayed at the top of the Print Server 2 interface, as shown in Figure 2-1 - Print Server ID . This is blank by default
<Print Server Port>	The TCP/IP port that the Print Server 2 will use. This is 2000 by default
<Binary Data Port>	The TCP/IP port that will be used for binary data. This is the Print Server Port number + 100 by default - please see the How to print images from the binary data guide for further details
<Configuration Folder>	The path to the configuration folder (see Configuration Menu) the Print Server 2 will use. This will be the previously used configuration folder (or the path set by the installer if the Print Server 2 has not been run previously) by default
<Start Minimised>	If this is "True", the Print Server 2 interface will start minimised in the system tray. The Print Server 2 interface will not start minimised by default

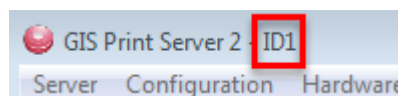


Figure 2-1: Print Server ID

For example, to launch Print Server 2 with an ID of "ID1", on Print Server Port 2001 and Binary Data Port 2101, using a configuration folder called "Default", and starting the Print Server maximised, the following command would be run:

"GIS Print Server 2.exe" "ID1,2001:2101,C:\ProgramData\Global Inkjet Systems\GIS Inkjet OS
2\Configurations\Default,False"

The parameters can be omitted from the command, which will cause Print Server 2 to start with the default values.



By default, Print Server launches using ports 2000 and 2100 but when launching Print Server from the command line or by using the Print Server Monitor tool, the port numbers can be manually selected.

Note that Print Server may fail to open or function correctly if another process or application is using a port number that is 5000 greater than that being used by Print Server. E.g., if Print Server is using ports 2000 and 2100, then ensure that no other process or application is using port 7000 and 7100.

3 Print Server Interface

The Print Server 2 interface is shown in **Figure 3-1 - Print Server 2 interface**.

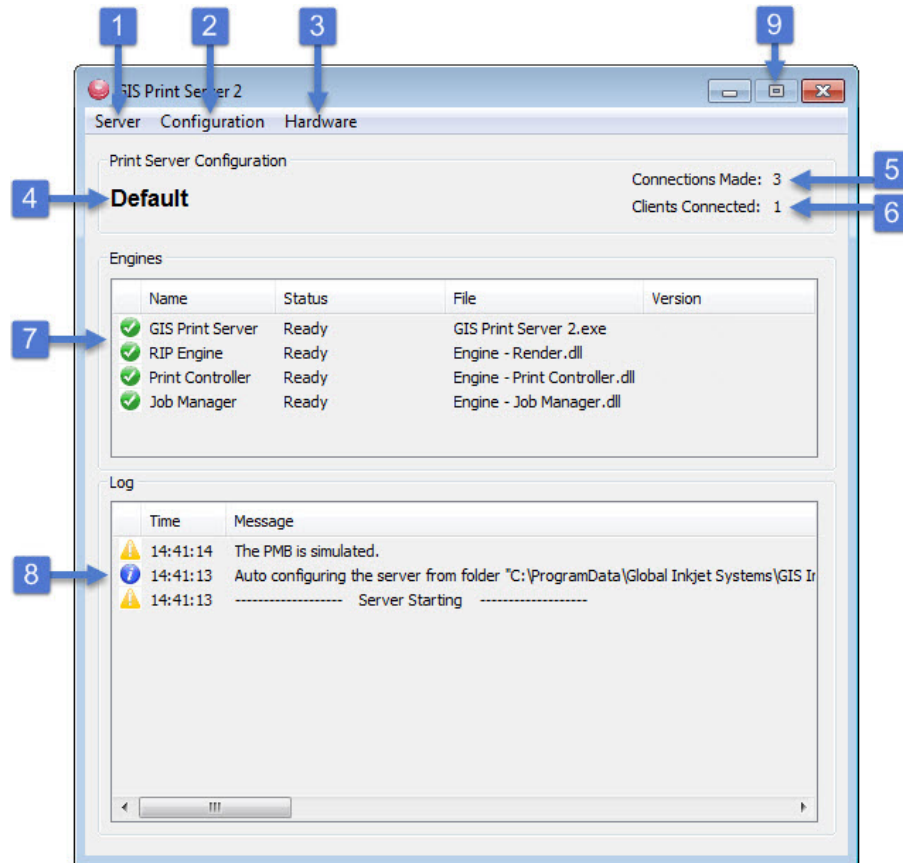


Figure 3-1: Print Server 2 interface

Table 1 - Print Server interface areas

Control	Name	Description
1	Server Menu	See section Server Menu
2	Configuration Menu	See section Configuration Menu
3	Hardware Menu	See section Hardware Menu
4	Configuration Folder	Currently used configuration folder (see section Configurations)
5	Connections Made	The number of TCP/IP connections made to the Print Server
6	Clients Connected	The number of TCP/IP clients currently connected to the Print Server
7	Engines List	See Engines Section
8	Message Log	See Log Section
9	Minimise / Close	Both active buttons will minimise the Print Server to the system tray. The Print Server won't shut down or change its behaviour in any way

3.1 Server Menu

The Server menu is shown in **Figure 3-2 Server Menu**.

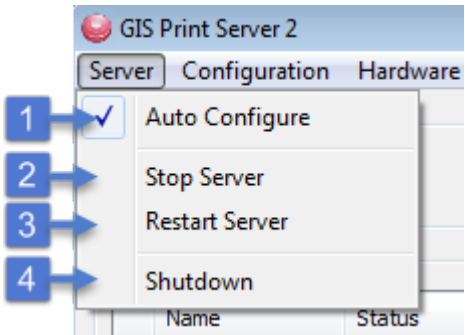


Figure 3-2: Server Menu

Table 2 - Server Menu items

Control	Name	Description
1	Auto Configure	If selected, on start up the Print Server will automatically load the appropriate configuration folder (see section Configurations) If not selected, the Print Server will not load any configuration when it starts up
2	Stop Server	Aborts the Print Server and all Engine Components Terminates all TCP/IP connections
3	Restart Server	Aborts the Print Server and all Engine Components and then reinitialises the Print Server and all Engine Components using the same configuration folder Terminates all TCP/IP connections
4	Shutdown	Aborts the Print Server and all Engine Components and then closes the Print Server

3.2 Configuration Menu

The Configuration menu is shown in **Figure 3-3: Configuration Menu**.

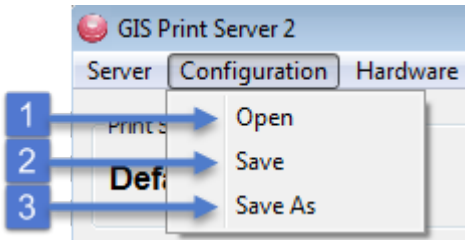


Figure 3-3: Configuration Menu

Table 3 -Configuration Menu items

Control	Name	Description
1	Open	See section Open Configuration .
2	Save	See section Save Configuration .
3	Save As	See section Save Configuration As .

3.2.1 Open Configuration

The open configuration option (control 1 in [Figure 3-3](#)) brings up the open configuration dialog, as shown in [Figure 3-4](#), which allows the path to the configuration folder to be selected.

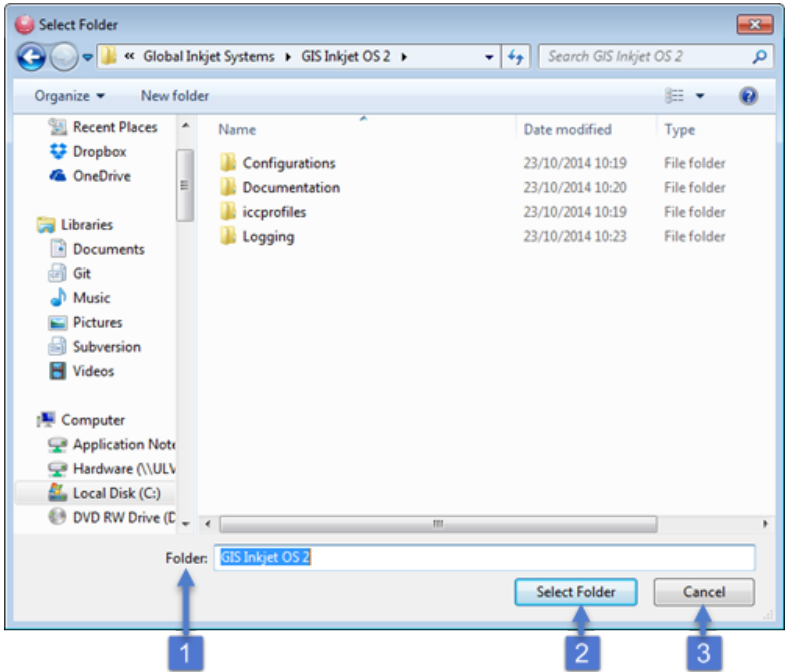


Figure 3-4: Open Configuration dialog

Table 4 - Open Configuration dialog items

Control	Name	Description
1	Folder name	The path to the configuration folder, please select a folder and not an individual file
2	Select	Confirm the configuration folder path
3	Cancel	Do not confirm the configuration folder path.

When the appropriate configuration folder is chosen, the Restart Print Server dialog, shown in [Figure 3-5](#), will be displayed. Clicking 'Yes' will automatically restart the Print Server, clicking 'No' will require the user to restart the Print Server manually by selecting: **Server>Restart Server** (Control 3 in [Figure 3-2](#)).

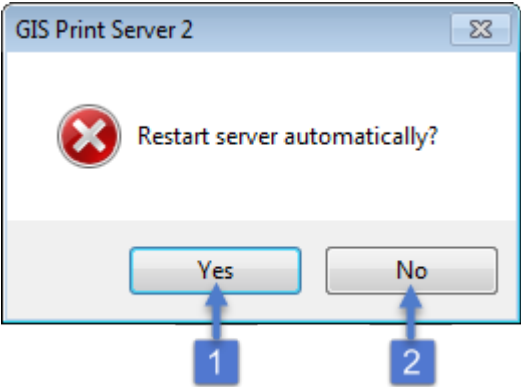


Figure 3-5: Restart Print Server dialog

Table 5 - Restart Print Server dialog items

Control	Name	Description
1	Yes	Confirm restart of Print Server using the new configuration folder
2	No	Cancel immediate restart of Print Server. The next time the Print Server starts, it will use the new configuration folder

3.2.2 Save Configuration

The save configuration option (control 2 in [Figure 3-3](#)) saves the configuration to the current configuration folder path. When this is selected, the change parameters dialog (see section [Changed Parameters Dialog](#)) is displayed.

3.2.3 Save Configuration As

The save configuration as option (control 3 in [Figure 3-3 - Configuration Menu](#)) brings up the save configuration as dialog, as shown in [Figure 3-6: Save Configuration dialog](#), which allows the new path to the configuration folder to be selected.

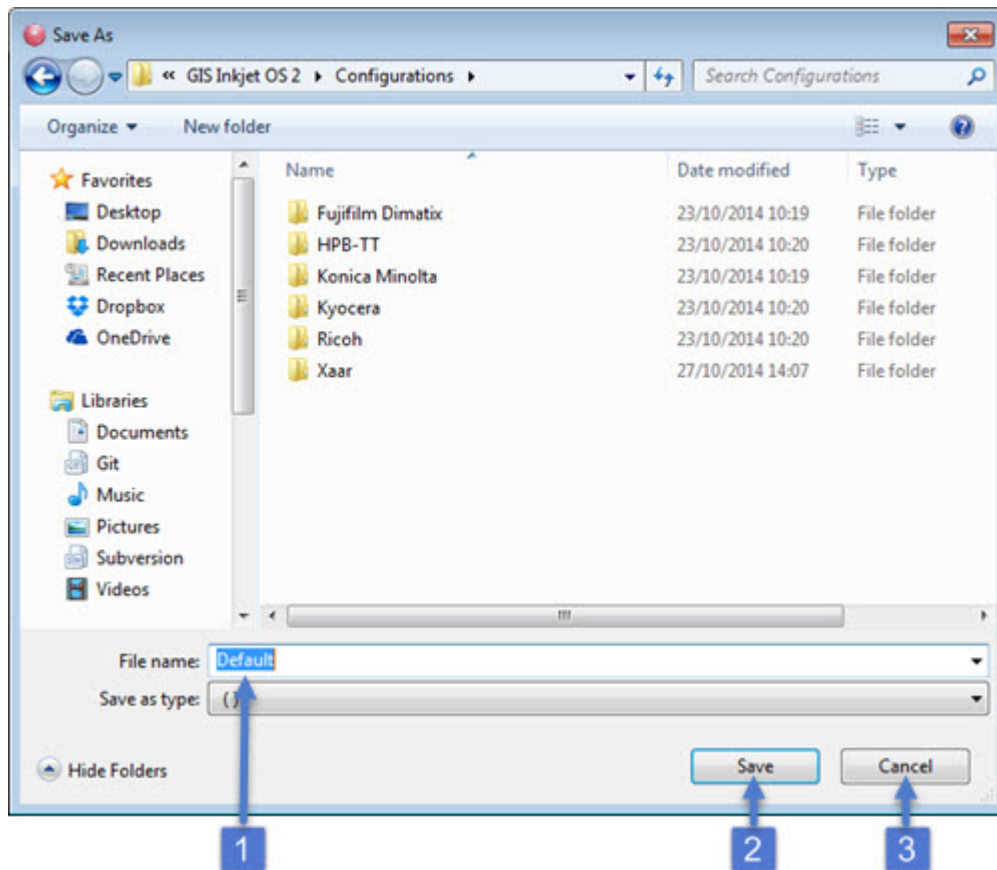


Figure 3-6: Save Configuration dialog

Table 6 - Save Configuration dialog items

Control	Name	Description
1	Configuration Folder Path	The new configuration folder path.
2	Save	Save the configuration to the selected path
3	Cancel	Do not save or change the configuration folder path

When the appropriate configuration folder is chosen the change parameters dialog (see section [Changed Parameters Dialog](#)) will be displayed.

3.2.4 Changed Parameters Dialog

The changed parameters dialog, which is shown in [Figure 3-7](#), details all of the parameters that have changed since the configuration was opened or last saved. The dialog provides options for controlling which of the changed parameters are saved and which ones are not.

The nodes and parameters for each Engine Component are separated, and any Engine Component or node that has no changes is not displayed. The nodes and parameters are arranged in a tree structure, which matches the structure of the nodes and parameter in the configuration file.

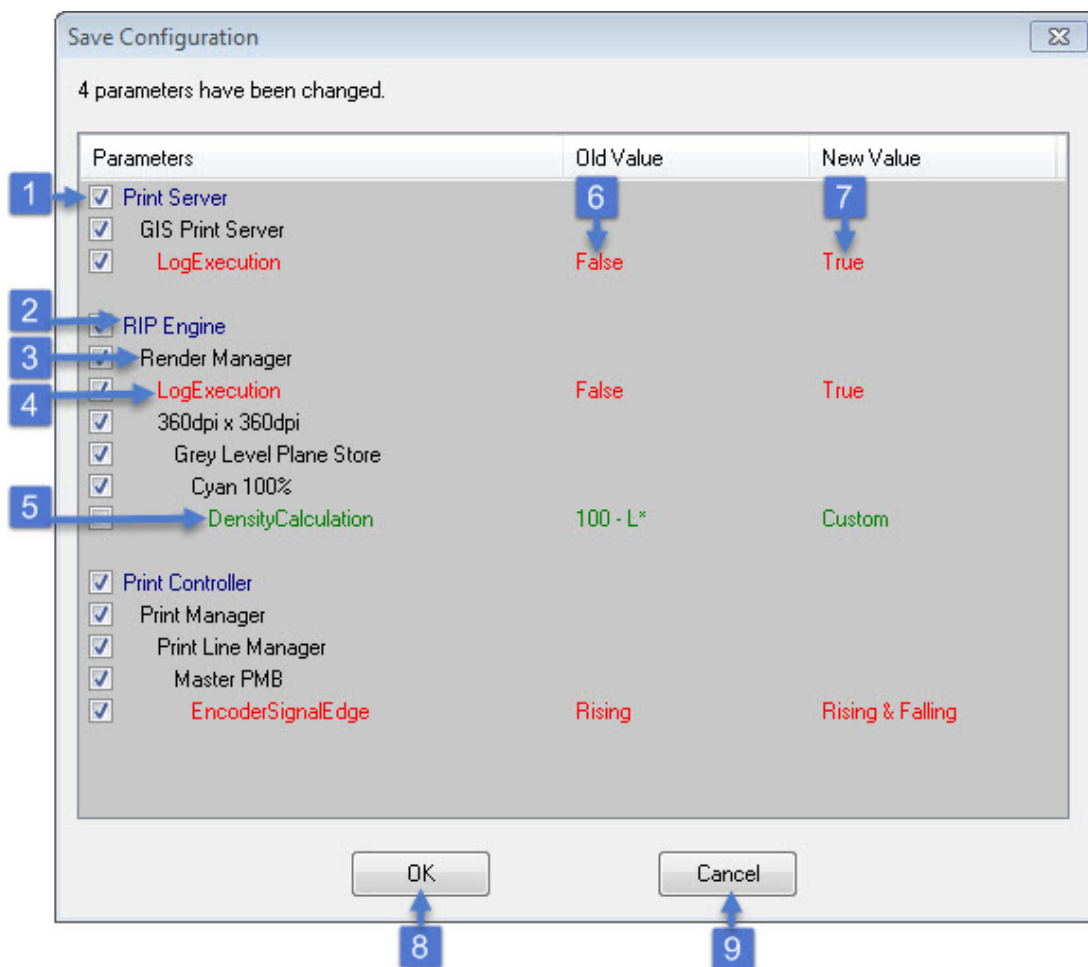


Figure 3-7: Changed Parameters dialog

Table 7 - Changed Parameters dialog items

Control	Name	Description
1	Save Selection Box	Only parameters with checked boxes will be saved
2	Blue Node Name	These are the names of the engines
3	Black Node Name	Tree nodes are black (see section Configuration Structure)
4	Red Parameter Name	Changed parameters to be saved to the configuration are red
5	Green Parameter Name	Changed parameters not to be saved to the configuration are green
6	Old Value	The old value of the parameter
7	New Value	The new value of the parameter to be saved
8	OK	Confirm the save of the changes to the configuration
9	Cancel	Cancel the save of the changes to the configuration

3.3 Hardware Menu

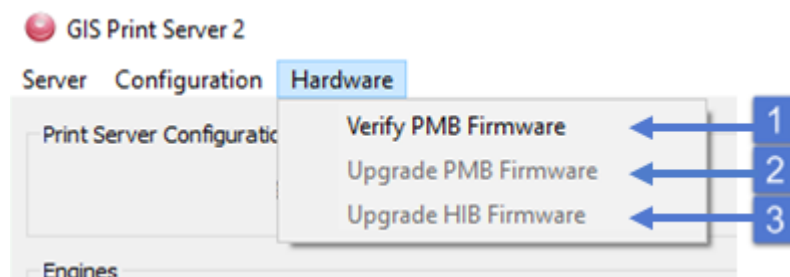


Figure 3-8: Hardware Menu

The Hardware menu has two options for upgrading the firmware on the GIS electronics. The Print Server will automatically detect the firmware version of the connected GIS hardware and will determine if it is compatible with the GIS software installed. If the version of the firmware is not correct then the Print Server will offer the user the option to upgrade the firmware.

Table 1 - Print Server interface areas

Control	Name	Description
1	Verify PMB Firmware	The Print Server will automatically detect the firmware version of the connected GIS PMB and determine if it is compatible with the GIS Software installed
2	Upgrade PMB Firmware	The Print Server will automatically detect the firmware version of the connected GIS PMB and determine if it is compatible with the GIS Software installed; it will upgrade remotely as required
3	Upgrade HIB Firmware	The Print Server will automatically detect the firmware version of the connected GIS HIB and determine if it is compatible with the GIS Software installed; it will upgrade remotely as required



Please be careful not to power down the electronics at any point during the upgrade. It is possible to damage the electronics if the firmware download is interrupted. If in doubt please contact GIS for detailed step-by-step instructions

When this option is used, there will be progress information in the Log window. The system can be used as normal once the progress indicator shows 100% and the hardware has been power cycled.

3.4 Engines Section

The engines section, shown in [Figure 3-9](#), lists the Print Server and all of the Engine Components that have been loaded in. Double clicking on an entry in the list will bring up the Tree Node Editor Dialog (see section [Tree Node Editor Dialog](#)) for this component.

	1	4	5	6	7
		Name	Status	File	Version
2	✓	GIS Print Server	Ready	GIS Print Server 2.exe	2.30000.0.0
	✓	RIP Engine	Ready	Engine - Render.dll	2.30000.0.0
3	!	Print Controller	ERROR Init Failed	Engine - Print Controller.dll	2.30000.0.0
	✓	Job Manager	Ready	Engine - Job Manager.dll	2.30000.0.0

Figure 3-9: Print Engines Information

Table 8 - Print Engines Information items

Control	Name	Description
1	Initialisation Status	This section shows if the print engine components have initialised successfully (see controls 2 and 3)
2	Successful Initialisation	Successful engine component initialisation shows a Green tick
3	Failed Initialisation	Failed engine component initialisation shows a Red exclamation mark
4	Engine Name	The engine component name
5	Engine Status	Current status of the engine component
6	Engine File	The name of the file that the component was created with
7	Engine Version	The software version of the engine component

3.5 Log Section

The log section, shown in **Figure 3-10**, lists all of the log messages produced by the Print Server and the Engine Components. The most recent log message is at the top of the list, and the number of messages displayed in the log is defined by a parameter in the Print Server.

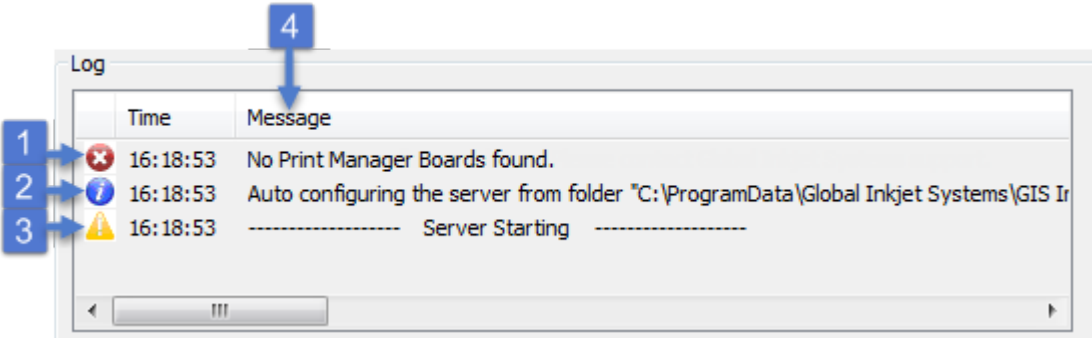


Figure 3-10: Log Information

Table 9 - Log Information items

Control	Name	Description
1	Error Message	Error messages are shown by a Red cross 
2	Information Message	Information messages are shown by a Blue i symbol 
3	Warning Message	Warning messages are shown by a Yellow exclamation mark 
4	Log Message Section	This section shows the log message

4 Configurations

4.1 Configuration Structure

The setup of Print Server 2 and all Engine Components are defined by 'nodes' and 'parameters'. The nodes represent a functional part of the system, such as a screener or a print head, and a parameter defines the value of a particular property associated with this node. Each node has a type, which defining its behaviour and available parameters. Each node is an individual component and so will behave independently, however many nodes can have the same type, for example, there may be many print heads in the system. The nodes are arranged in a tree structure, which means that each node can have a parent or children. The children of a node are functional parts which contribute to forming the parent node. For example, a print head node may have print nozzle group child nodes, as print nozzle groups form part of a print head. An example of a tree structure is shown in

Figure 4-1.

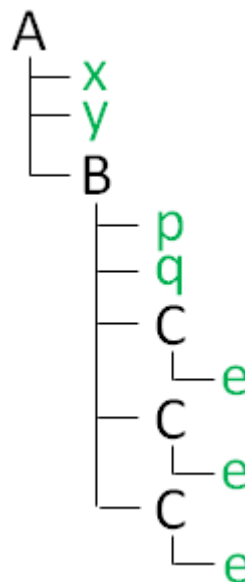


Figure 4-1: Tree example

The tree example in **Figure 4-1** shows a tree which contains 5 nodes (shown in black). Each node has associated parameters (shown in green). The node names indicate the type of the node, so while there are 5 nodes, there are only 3 different types of node. Node B is a child of node A, meaning that the type B contributes to how A works. Node B has 3 children, however all are of the same type so they will perform similar but separate actions. Node A has no parent node, and so it is called the 'Root' node.

4.1.1 Parameter Types

Every parameter associated with a node has a type. The type of a parameter defines what sort of information it can hold, and other properties that may or may not be associated with it.

Boolean

A Boolean parameter can only take the value 'True' or 'False'.

Integer

An integer parameter can take any whole number. Integer parameters have minimum and maximum values, which define the range of values that the parameter can be set to.

Decimal

A decimal parameter can take any decimal value (up to 8 decimal places). Decimal parameters have minimum and maximum values, which define the range of values that the parameter can be set to.

Text

A text parameter can have any text value.

Selection

A selection parameter has a pre-defined list of values, and the parameter can only be set to one of these values.

4.2 Configuration Files and Folders

The setup of the Print Server and Engine Components trees are specified by configuration files. These are XML files which define the structure of all nodes and the values of every parameter for every node in the tree. The Print Server works with configuration folders, which contain a number of these configuration files.

The Print Server will always be initialised using the configuration file "Print Server.xml" inside the configuration folder. The Engine Components which are to be included are specified by the Print Server. The Print Server has a node for each Engine Component, with parameters defining the name of the engine and the DLL file for the engine. The engine will then be created using this DLL file, and it will be initialised using the configuration file "<Engine Name>.xml" inside the configuration folder.

5 Configuring the Print Server

5.1 Configuration Overview

Configuring the Print Server involves the following steps:

Step 1: [Create the Configuration folder](#)

Step 2: [Copy the Security File](#)

Step 3: [Load the Configuration folder](#)

Step 4: [Link to a Security File](#)

Step 5: [Gather the PMB Serial Number](#)

Step 6: [Link to the connected PMB](#)

Step 1: Create the Configuration Folder

Add all Configuration files to a folder of your choice. We recommend creating a 'Config' folder inside the installation directory:

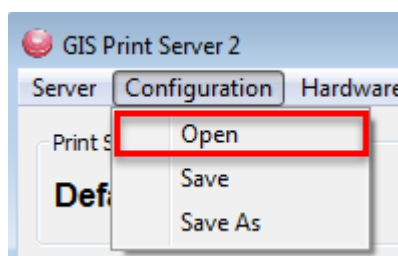
C:\ProgramData\Global Inkjet Systems\GIS Inkjet OS 2. In this example, the folder is called 'Config'.

Step 2: Copy the Security File

Copy the security file into the installation directory, overwriting any existing security files if prompted.

Step 3: Load the Configuration Folder

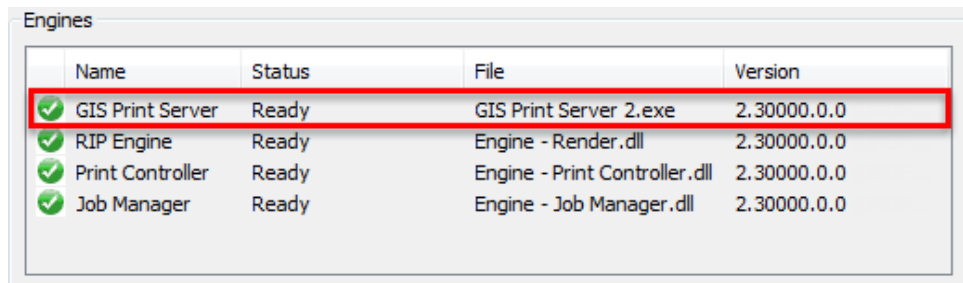
- 1 Open the GIS Print Server
- 2 Click on Configuration > Open from the menu bar



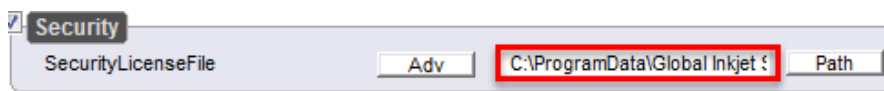
- 3 Navigate to the Config folder created in Step 1 and press the Select button

Step 4: [Link to the Security File](#)

- 1 Double click on the GIS Print Server Engine Component



- 2 Add the path to the security file copied in Step 2 into the **SecurityLicenseFile** parameter

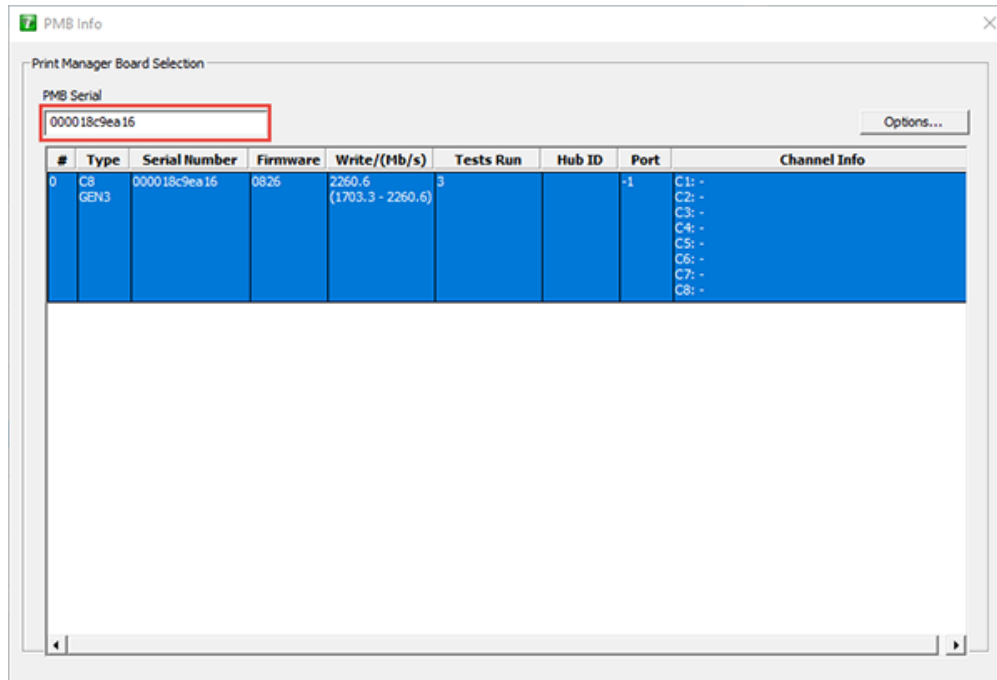


Step 5: [Gather the PMB Serial Number](#)



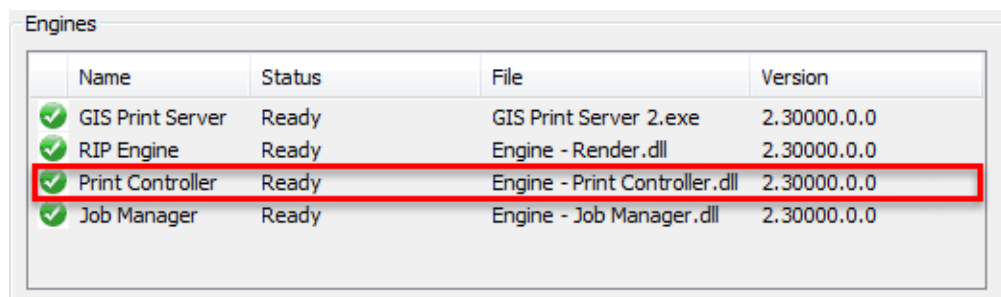
Note that Print Server should be shut down before accessing the PMB information in Step 5 if set up with the default settings; restart before Step 6. See [Launching Through the Command Line on page 5](#) for details on port settings

- 1 Run the GIS utility - PMB Info application (located in the installation directory)
- 2 The list of all connected PMBs will be displayed and the LEDs on the front of all of the PMBs will turn red
- 3 Click on a PMB in the list. The LEDs on the front of the PMB this list entry represents will turn green
- 4 Find the desired PMB in the list and record the serial number, which is shown in the 'PMB Serial' box

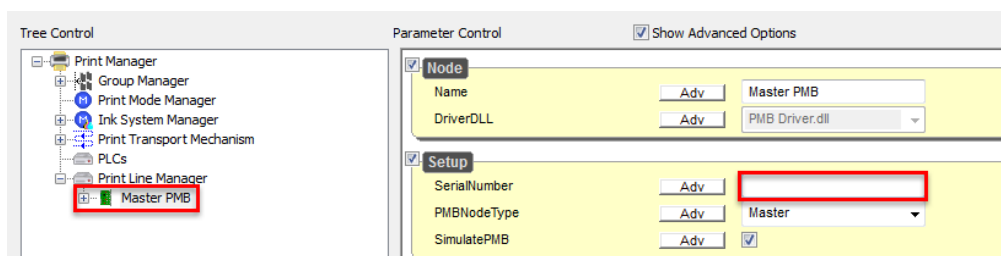


Step 6: Link to the Connected PMB

- 1 Double click on the Print Controller Engine Component

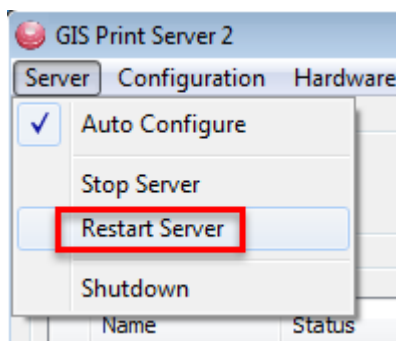


- 2 In the Tree Control, expand Print Line Manager and click on the PMB node. The tree structure shown may differ, depending on the contents of the configuration file



- 3 Enter the serial number recorded in step 6 into the SerialNumber parameter

- 4 Close the Tree Editor
- 5 Click on Server > Restart Server from the menu bar



- 6 Wait until the restart has fully completed

6 Tree Node Editor Dialog

The Tree Node Editor dialog, shown in [Figure 6-1](#), is used to change the structure of a tree or edit the value of parameters associated with nodes in the tree (see section [Configuration Structure](#)).

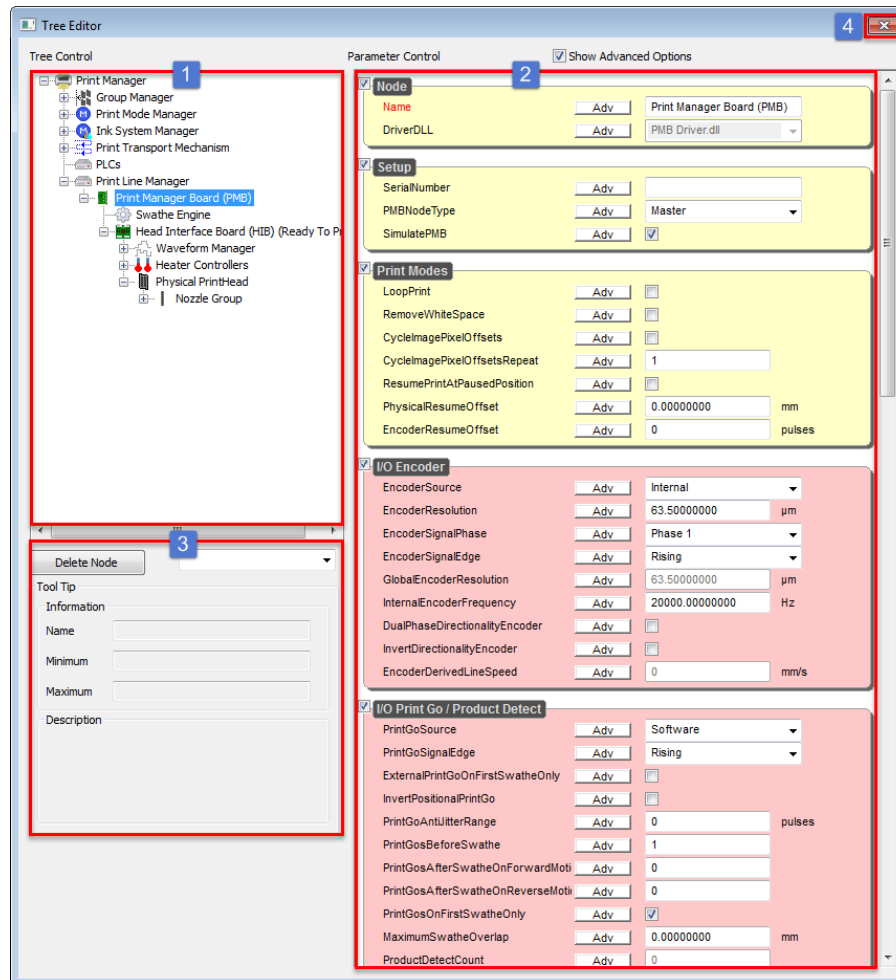


Figure 6-1: Tree Node Editor dialog

Table 10 - Tree Node Editor dialog items

Control	Name	Description
1	Tree Control	See section Tree Control
2	Parameter Control	See section Parameter Control
3	Parameter Information	See section Parameter Information
4	Close	Close the Tree Node Editor Dialog. This will not cause any parameters to be saved to the configuration file

6.1 Tree Control

The tree control section of the Tree Node Editor Dialog, shown in [Figure 6-2](#), is used to view and adjust the structure of the nodes which make up the configuration. The nodes are displayed in a tree format (see section [Configuration Structure](#)).

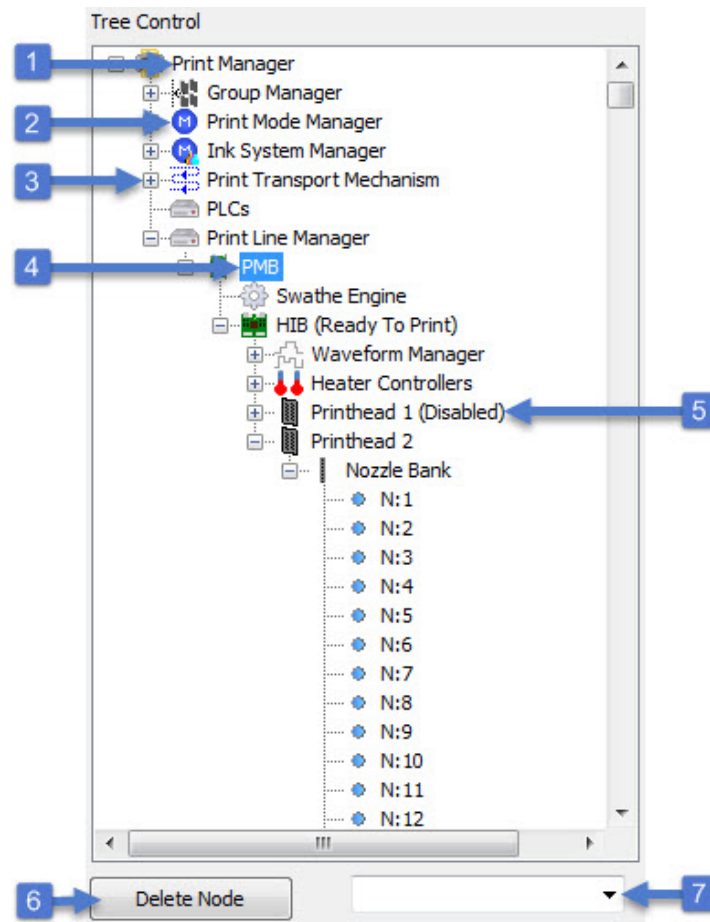


Figure 6-2: Tree Control

Table 11 - Tree Control items

Control	Name	Description
1	Node Name	The name of the node
2	Node Icon	An icon representing the type of the node
3	Expand Node	Clicking on this button will cause all of the associated node's children to be displayed. Clicking on it again will hide all of the child nodes
4	Selected Node	A node with a solid blue background is a currently selected node
5	Disabled Node	If a node has a parameter <code>IsEnabled</code> and this is set to <code>False</code> , the node will be marked as Disabled
6	Delete Node	Delete the currently selected node and all of its children
7	Add Child Node	See section Add Child Node

6.1.1 Add Child Node

To add a child node to the currently selected node, the add node selection box (control 7 in [Figure 6-2](#)) should be used. This box lists all available node types that the child node can take, as shown in [Figure 6-3](#). The appropriate node type should be selected to add a node of this type.

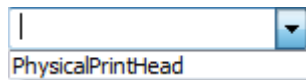


Figure 6-3: The add child node selection box

Some child nodes will auto-populate themselves when added, which may result in child nodes being automatically added to the new child node.

Adding a new HIB Node

To add a new HIB Node:

- 1 Highlight the PMB Node in the Tree Control
- 2 Select HIB from the dropdown menu - a new HIB (Unknown) will be added
- 3 Highlight the HIB (Unknown) in the Tree Control
- 4 In the Node > DriverDLL Parameter Control, select the driver for the HIB. The Waveform Manager and Heater Controller nodes will be automatically created in the Tree Control
- 5 In the Node Parameter Control, give a name to your HIB, we recommend naming the HIB to reflect the connector or ink being used

Adding a new Printhead Node

To add a new Printhead Node to a HIB:

- 1 Highlight the HIB Node in the Tree Control
- 2 Select PhysicalPrintHead from the dropdown menu - a new Printhead will be added
- 3 Highlight the Printhead in the Tree Control
- 4 In the Node Parameter Control, give a name to your Printhead, we recommend naming the Printhead to reflect the connector or ink being used
- 5 In the Communications Parameter Control, select the ConnectorIndexOnHIB - this should be the Connector your Printhead is connected to on the HIB

6.2 Parameter Control

The parameter control section of the Tree Node Editor Dialog, shown in [Figure 6-4](#), is used to view and adjust the values of the parameters for the currently selected node. The parameters are organised into sections, with coloured backgrounds to make it easier to locate a parameter.

Parameter Control ☒ Show Advanced Options

1

Node

Name

Adv

Print Manager Board (PMB)

11

DriverDLL

Adv

PMB Driver.dll

3

Setup

SerialNumber

Adv

PMBNodeType

Adv

Master

SimulatePMB

Adv

☒

4

Print Modes

LoopPrint

Adv

☐

7

RemoveWhiteSpace

Adv

☒

8

CycleImagePixelOffsets

Adv

☐

CycleImagePixelOffsetsRepeat

Adv

1

ResumePrintAtPausedPosition

Adv

☐

PhysicalResumeOffset

Adv

0.00000000

9

EncoderResumeOffset

Adv

0

10

16

I/O Encoder

EncoderSource

Adv

Internal

12

EncoderResolution

Adv

63.50000000

µm

14

EncoderSignalPhase

Adv

The distance between two rising edges of #

EncoderSignalEdge

Adv

Rising

GlobalEncoderResolution

Adv

63.50000000

µm

13

InternalEncoderFrequency

Adv

20000.00000000

Hz

DualPhaseDirectionalityEncoder

Adv

☐

InvertDirectionalityEncoder

Adv

☐

EncoderDerivedLineSpeed

Adv

0

mm/s

5

I/O Print Go / Product Detect

Print Mode

GreyLevelAssociation

Adv

-1

GreyLevelReduction

Adv

0

MirrorSwathe

Adv

☐

InvertToggle

Adv

☐

MultipassMapFilename

Adv

Path

15

Figure 6-4: Parameter control items

Table 12 - Parameter control items

Control	Name	Description
1	Parameter Name	The name of the parameter
2	Changed Parameter	Parameter names in red indicate that the value of the parameter has changed since the configuration was opened or last saved
3	Parameter Section Name	The name of the parameter section
4	Hide Section Button	The button which controls whether the associated parameter section is hidden or not
5	Hidden Section	A parameter section which has been hidden
6	Show Advanced Options	The advanced parameters will only be displayed if this button is checked
7	False Boolean Parameter	A boolean parameter whose value has been set to "False" (see section Boolean)
8	True Boolean Parameter	A boolean parameter whose value has been set to "True" (see section Boolean)
9	Decimal Parameter	A decimal parameter (see section Decimal)
10	Integer Parameter	An integer parameter (see section Integer)
11	Text Parameter	A text parameter (see section Text)
12	Selection Parameter	A selection parameter (see section Selection)
13	Parameter Unit	The unit for this parameter
14	Parameter Tooltip	If the cursor hovers over the parameter for a few seconds, a tooltip describing the parameter will be displayed
15	Path Button	See section Path Button
16	Advanced Parameter Settings	This displays the Advanced Parameter Settings dialog (see section Advanced Parameter Settings Dialog) for the associated parameter

The Print Server will be able to use the new parameter value entered into a value box as soon as any other part of the Tree Node Editor Dialog is clicked.

6.2.1 Path Button

Some parameters require the value to be a filename or file path. If this is the case, the unit will be replaced with the Path button (control 15 in [Figure 6-4](#)). When this button is clicked, the folder selection dialog, shown in [Figure 6-5](#), will be displayed.

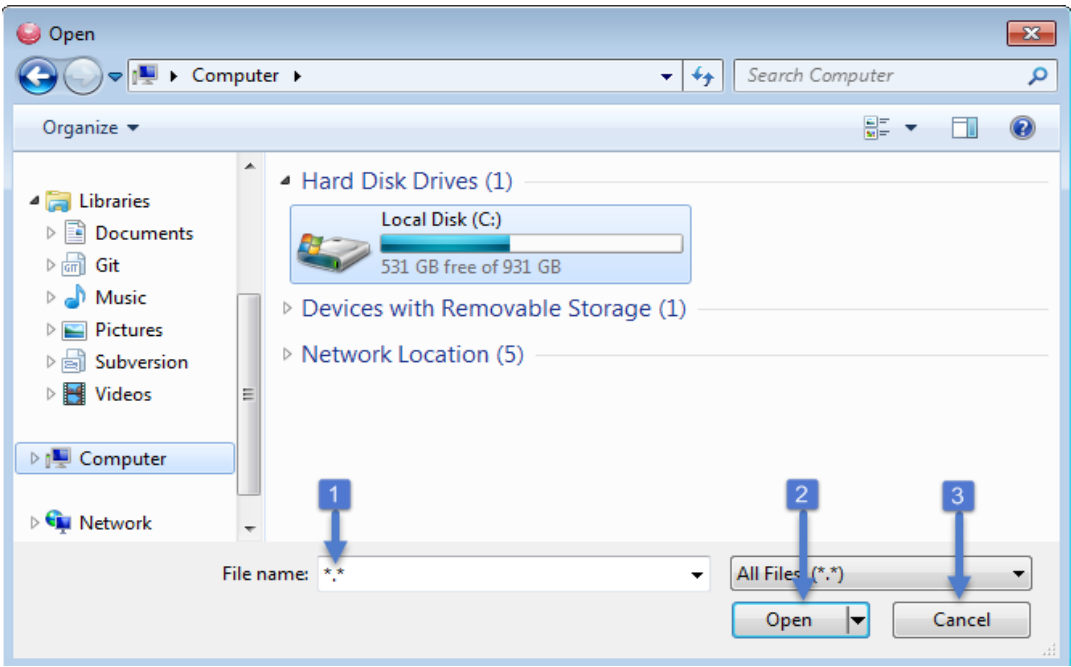


Figure 6-5: Choose Path dialog

Table 13 - Path dialog items

Control	Name	Description
1	File Path or Name	The file path or name
2	Open	Use the current folder path (as shown by control 1) as the parameter value
3	Cancel	Do not change the parameter value

6.2.2 Parameter Information

When a parameter value box is clicked, the parameter information section, shown in [Figure 6-6](#), will display information about this parameter.

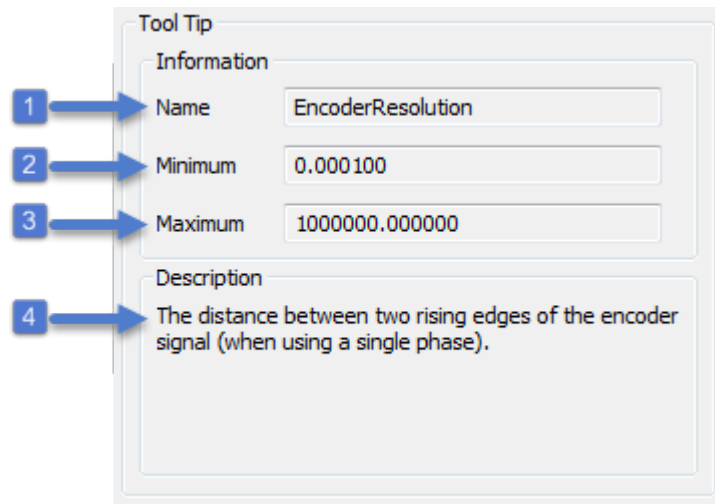


Figure 6-6: Parameter Information

Table 15 - Parameter Information items

Control	Name	Description
1	Parameter Name	The name of the parameter
2	Minimum Value	The minimum value this parameter can be set to. This only applies to parameters types that have ranges (see section Parameter Types)
3	Maximum Value	The maximum value this parameter can be set to. This only applies to parameters types that have ranges (see section Parameter Types)
4	Description	A description of the parameter

6.3 Advanced Parameter Settings Dialog

When the Adv button in the Tree Node Editor Dialog (control 16 in [Figure 6-4](#)) is clicked, the Advanced Parameter Settings dialog, shown in [Figure 6-7](#), is displayed.

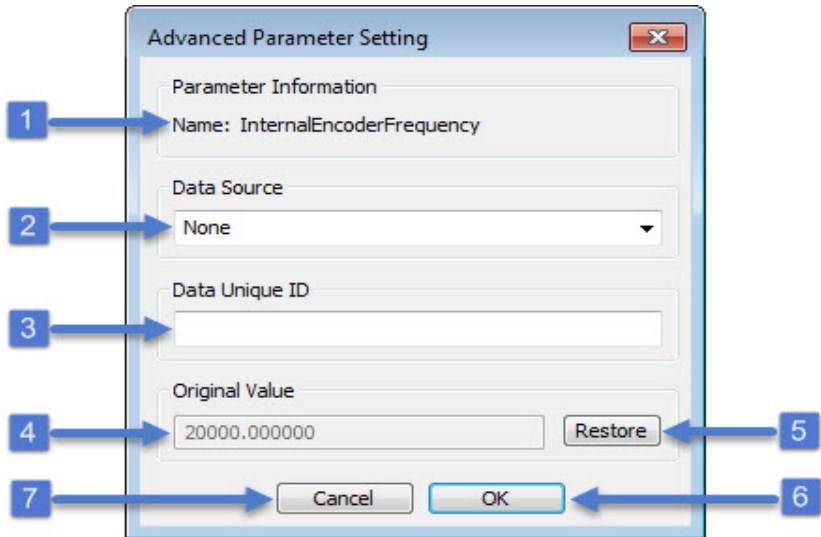


Figure 6-7: Advanced Parameter Settings dialog

Table 16 - Advanced Parameter Setting dialog items

Control	Name	Description
1	Parameter Name	The name of the parameter
2	Data Source	The data source of this parameter. This is used in variable data printing (see the <i>Layout User Guide</i>)
3	Data Unique ID	The unique ID of this parameter. This is used in variable data printing (see the <i>Layout User Guide</i>) and TCP/IP commands (see the <i>Print Server Communication User Guide 2.0</i>)
4	Original Parameter Value	The value of the parameter the last time the configuration was opened or saved
5	Restore Original Parameter Value	Change the value of the parameter to the original value (shown in control 4)
6	OK	Save the changes made and close the dialog
7	Cancel	Cancel the changes made and close the dialog

7 Troubleshooting

7.1 Space Requests Have Been Missed

The Print Server constantly monitors the Status Reports coming back from the PMB.

These Status Reports are sent back from the PMB whenever it completes a label or a Swathe Section.

Each Status Report has an ID number that increments by one every time. If the Print Server notices that there is a jump in the Status Report ID numbers, then it reports the error (where x is the number by which the sequence has jumped up by);

"x Space requests have been missed"

This message indicates that the CPU did not process a USB response (from the PMB) and happens when the CPU is temporarily overloaded. If you only see a few of these now and again then it is nothing to worry about as the system will handle it. This may have been caused by virus checkers, an application starting, moving windows or almost any other action.

However if you see large numbers of these messages then it might be a symptom of a larger problem.

7.2 Print Server Parameters

There are two parameters in the Print Server that allow you to control for how long and how frequently you want the Print Server to listen for Status Reports from the PMB. These can be found in the Threading node in the Swathe Engine as shown in [Figure 7-1](#).

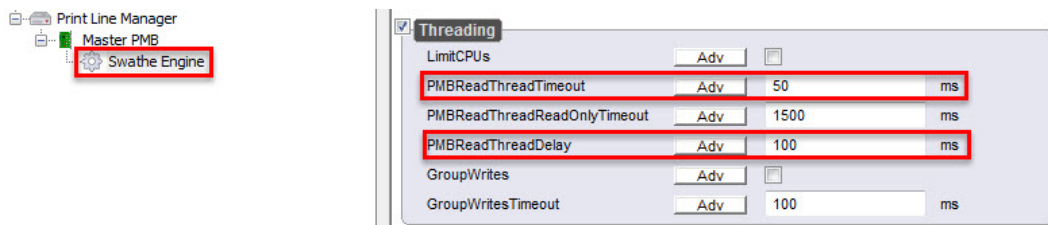


Figure 7-1: Adjusting PMB threading settings in the Print Server

PMBReadThreadTimeout

The PMBReadThreadTimeout indicates the amount of time that the PC listens for a Status Report to be sent by the PMB. If no report is seen in this time it will delay for the period according to the PMBReadThreadDelay.

PMBReadThreadDelay

If there are no reports from the PMB within the PMBReadThreadTimeout period, the Print Server will stop listening for Status Reports for the Delay time before starting listening again.

7.3 An Example

Figure 7-2 shows four Status Reports being sent from the PMB to the Print Server, our example is using the PMB Threading settings from Figure 7-1.

The Print Server is set to listen (PMBReadThreadTimeout) for a Status Report every 50ms and if no Status Report is captured in this period, it will stop listening and delay (PMBReadThreadDelay) for 100ms.

- Status Report 1 and Status Report 2 are sent by the PMB whilst the Print Server is listening, so both are acknowledged and processed by the Print Server.
- Status Report 3 is sent by the PMB whilst the Print Server is not listening, thus is missed.
- Status Report 4 is sent by the PMB whilst the Print Server is listening, so is acknowledged and processed by the Print Server. However, because the Print Server is expecting Status Report 3, an error will be generated to state how many Status Reports have been missed:
"1 space request missing".

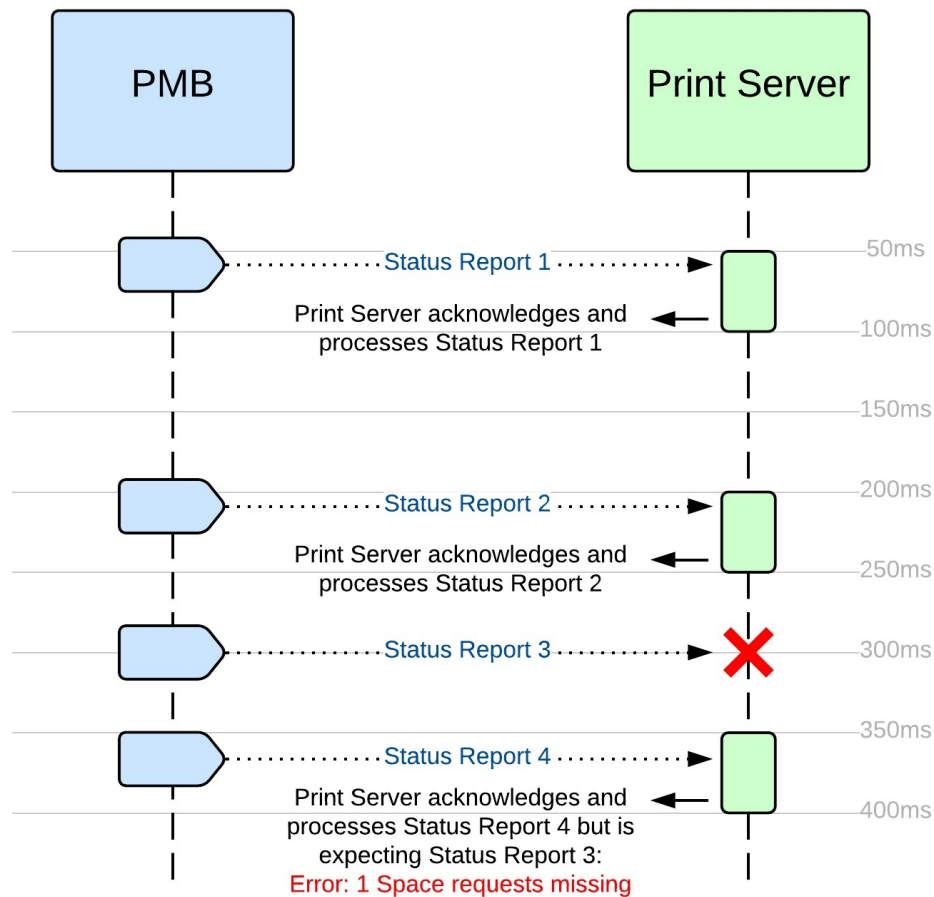


Figure 7-2: Example of missing space requests

Recommended settings

There are no ideal settings, but they will vary depending on the PC specifications, the USB transmission speeds and Print speeds.

If too many Space errors are being seen, try the following:

- 1 Decrease the value of PMBReadThreadDelay

This will reduce the time the Print Server is not monitoring for Status Reports. Decreasing this amount will increase the USB bandwidth dedicated to listening for Status Reports. If too much bandwidth is used up this will affect the amount of data available for printing.

- 2 Increasing the value of PMBReadThreadTimeout

This will increase the time the Print Server is monitoring for Status Reports.
Increasing this amount will decrease the amount of time where the USB bandwidth is not being used for listening for Status Reports. If too much bandwidth is used up this will affect the amount of data available for printing.

8 Contact / Support - Information

We have a dedicated team of in-house experts who pride themselves on providing you with world-class support.

Based in Cambridge, UK, we also have support offices in China and Japan. We provide:

- Worldwide coverage
- On-site support
- Training
- Software support
- Remote support sessions

If you require further information on GIS software and electronics products or ink system components, then please contact us by any of the following methods:

GIS Support Method	Contact
Log a support request / enquiry	GISsupport@globalinkjetsystems.com
Learn more about GIS and other contact information	www.globalinkjetsystems.com
Feedback	suggestions@globalinkjetsystems.com



9 Revision History

Rev.	Date	Author	Comment
1.0	11/12/14	ML	Flare Initial Version
1.1	05/01/18	NC	Minor amends to Section 2.1, 3.3 and 5.1 and updated styling throughout
1.2	16/12/20	NC	Updated Disclaimer and added Contact Support sections
1.3	12/01/22	AS	Updated section 2.1 Launching Through the Command Line
1.4	22/02/22	NC	Updated Section 5 Configuring the Print Server